

ASSESSMENT

Hardware Components & System Assembly

Section A — Concept Applications

S1. Selecting compatible components SCENARIO

You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.

QUESTION

Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

ANSWER:-

I would first choose a motherboard that fits the case, such as ATX, Micro-ATX, or MiniITX. I would match the CPU socket to the motherboard socket and check the motherboard manufacturer's CPU support list and BIOS requirements. I would select RAM with the correct generation, such as DDR4 or DDR5, and check its capacity, speed, voltage, and the motherboard's maximum supported memory. For storage, I would check whether the board has the required SATA ports or an M.2 slot supporting the selected NVMe drive. I would also check physical dimensions, cooling, expansion slots, and PSU connectors. Finally, I would confirm the parts using the motherboard manual, manufacturer compatibility lists, BIOS support information, case specifications, and a reliable PC-part compatibility checker.

S2. HDD, SSD, and NVMe storage

SCENARIO

You are advising a client whose employees mainly run office and productivity applications on their workstations.

QUESTION

Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

ANSWER:-

An HDD stores data on spinning magnetic disks and uses moving parts. It offers large capacity at low cost but is slower, noisier, and more easily affected by movement. A SATA SSD stores data in flash memory, has no moving parts, and is much faster and quieter than an HDD. An NVMe SSD also uses flash memory but communicates through PCIe, usually through an M.2 slot, so it normally provides lower access time and higher transfer speed than SATA. For office workstations I recommend an SSD, preferably NVMe when the price and motherboard support are suitable. It provides faster booting, quicker application loading, better responsiveness, low noise, and better resistance to shock. A SATA SSD is still an excellent lower-cost option; an HDD may be used as secondary bulk storage or backup.

S3. GPU and cooling

SCENARIO

You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.

QUESTION

Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

ANSWER:-

The GPU processes graphics, images, video, 3D views, and visual effects. A suitable GPU makes design software smoother and can reduce graphics work handled by the CPU. The GPU and CPU produce heat during long periods of use. The cooling system, including heatsinks, thermal paste, CPU/GPU fans, and case airflow, removes this heat. Good cooling prevents thermal throttling, crashes, freezes, unexpected shutdowns, and possible component damage. I would choose hardware that meets the software requirements, ensure the cooler is firmly attached, connect all fan headers, provide good intake and exhaust airflow, and keep dust away from vents.

S4. Hardware compatibility

SCENARIO

You are assembling a PC from components sourced from three different vendors.

QUESTION

Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

ANSWER:-

Compatibility matters because components must fit physically, use matching electrical connections, and communicate using supported standards. Possible problems include a CPU that does not fit the motherboard socket, a CPU unsupported by the BIOS, DDR4 RAM placed in a DDR5 board, an M.2 drive unsupported by the slot, a GPU too large for the case, an underpowered PSU, missing power connectors, an unsuitable CPU cooler, or a motherboard that does not fit the case. These problems can cause failure to boot, missing hardware in BIOS, crashes, overheating, or permanent damage. I would check manuals, dimensions, socket and chipset support, RAM type, storage interface, PSU requirements, and compatibility lists before assembly.

S5. Choosing and connecting the PSU

SCENARIO

You are finalizing the build and need to connect the PSU to the motherboard, GPU, and storage devices before first boot.

QUESTION

Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

ANSWER:-

I would add the expected power use of the CPU, GPU, drives, fans, and other devices, then choose a quality PSU with approximately 20–30% spare capacity. I would check the PSU efficiency rating, protections, trusted manufacturer, and required connectors. Before powering on, I would unplug the mains cable, switch off the PSU, use ESD precautions, and check that the motherboard is mounted on the correct standoffs. I would connect the 24-pin motherboard cable, 4/8-pin CPU cable, GPU PCIe cable, and SATA power cables. I would connect SATA data cables where required, connect the CPU cooler to CPU_FAN, and check the RAM, GPU, and storage seating. I would inspect for loose screws or wires that could cause a short circuit. Modular cables must belong to the exact PSU model, and connectors must never be forced.

S6. POST checks and failure procedure

SCENARIO

You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.

QUESTION

Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

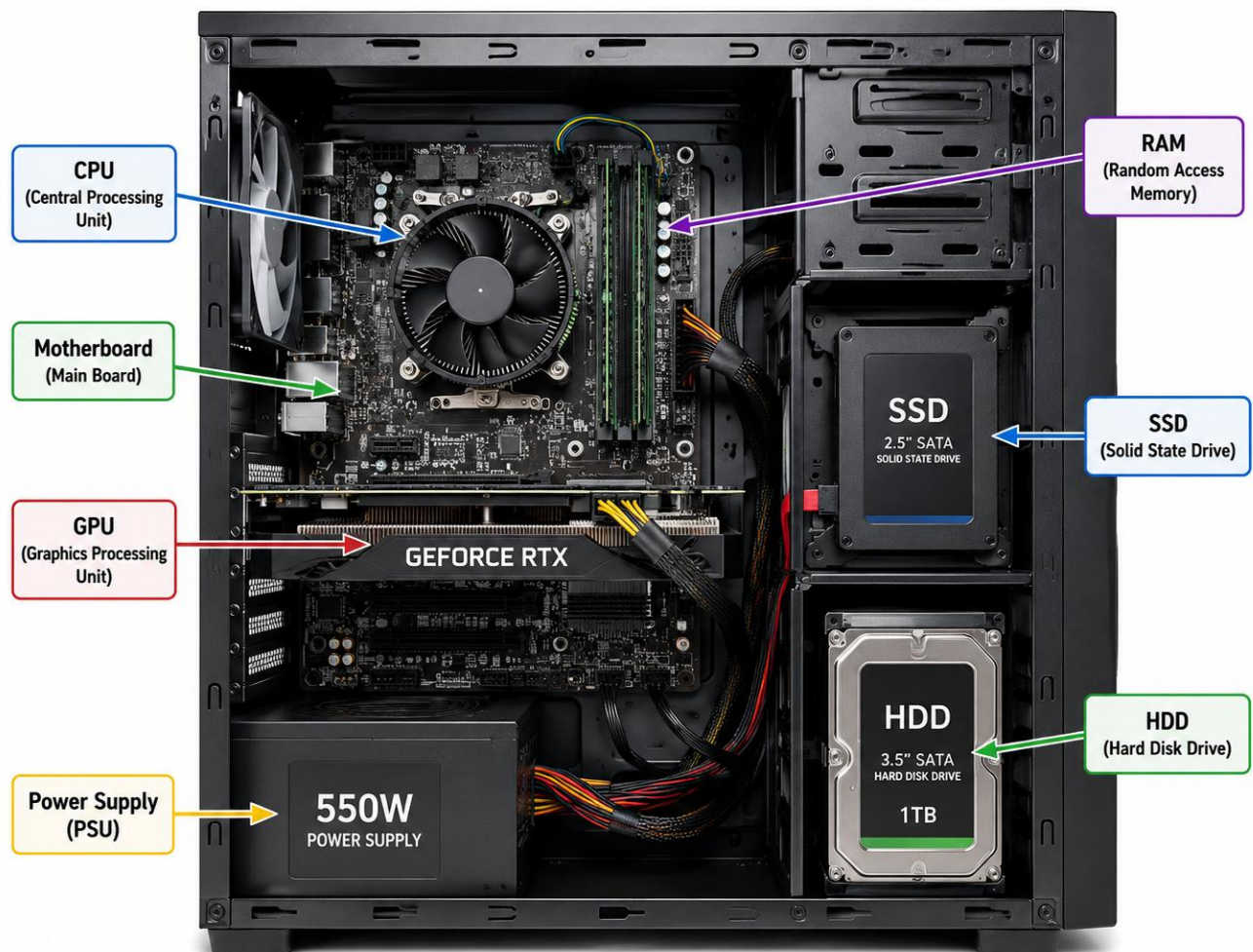
ANSWER:-

I would check that the system powers on, fans operate normally, the monitor shows the manufacturer logo or POST screen, and there are no error beeps or diagnostic LED errors. I would enter BIOS and confirm the CPU model, correct RAM capacity, storage device, and safe temperatures. I would then confirm that the operating system starts. If POST failed, I would switch off and unplug the PC, check the 24-pin and CPU power cables, reseal RAM and test one module, reseal the GPU and its power cable, inspect diagnostic LEDs or beep codes, check CPU installation and compatibility, clear CMOS if appropriate, and test minimum hardware. I would inspect for bent pins, loose screws, or required BIOS updates and test suspected parts one at a time.

Section B — Practical Tasks

Task 1: Component Identification & Functions

DESKTOP PC – Labeled Diagram



Labelled Component List:

1. Motherboard – Main circuit board that connects all hardware components.



MOTHERBOARD

1. CPU – Performs calculations and executes program instructions.



CPU

2. RAM – Temporary memory used by active applications.



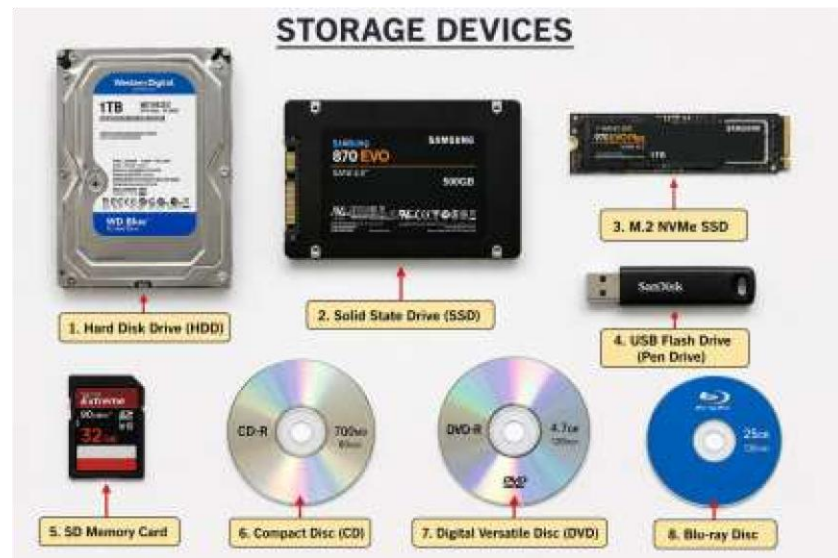
RAM

3. GPU – Processes graphics and video output.



GPU

4. Storage Device – Stores operating system, software, and user files.



STORAGE DEVICES

5. Cooling System – Removes heat and maintains safe operating temperatures.



COOLING FAN

Task 2: Motherboard Ports & Socket Identification



CPU Socket & Compatibility

- Socket: Intel LGA 1700
- Compatible CPUs: Intel 12th, 13th, and 14th Gen Core (e.g., i9-12900K, i7-13700K).

Expansion Slots

- PCIe 5.0 x16: Primary slot for dedicated GPU.
- PCIe 4.0 x16 (x4 mode): For expansion cards (sound, capture cards).
- M.2 NVMe (Gen4x4): For high-speed NVMe SSDs.
- M.2 Wi-Fi/BT: For wireless networking cards.

Internal Headers & Ports

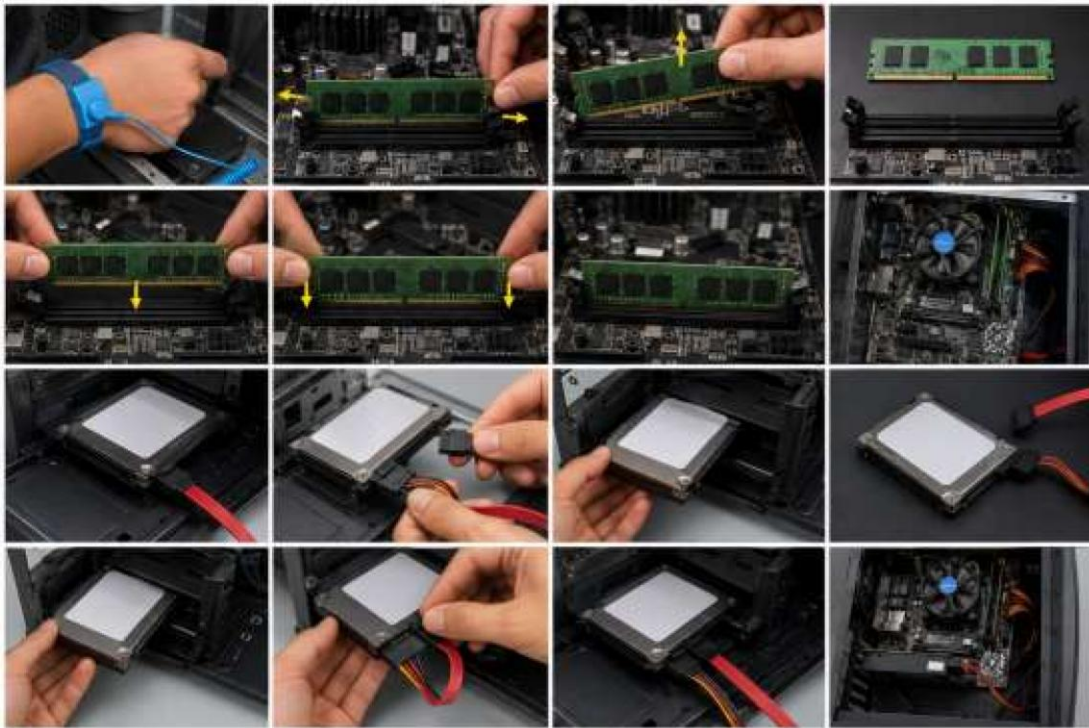
- 24-pin ATX Power: Main motherboard power.
- 8+4-pin EPS: Dedicated CPU power.
- SATA 6 GB/s: Storage ports for 2.5" SSDs / 3.5" HDDs.
- USB Headers: Front-panel USB Type-C, USB 3.2, and USB 2.0.
- Audio Header: Front-panel mic/headphone jacks.
- Fan & Pump Headers: Power for CPU cooler, AIO pump, and fans.
- ARGB Headers: Lighting control for RGB fans/strips.

Rear I/O Panel

- HDMI / DisplayPort: Video output for integrated graphics
- .USB Type-A & Type-C: High-speed peripheral connection.

- 2.5 Gb Ethernet: Wired network connection.
- Wi-Fi Antenna Connectors: Wireless network antenna jacks.
- Audio Jacks / SPDIF: Sound output for speakers/headsets.
- BIOS Flashback / Clear CMOS: BIOS updates and settings reset.

Task 3: RAM & Storage Installation



RAM Installation

1. Shut down the computer and unplug the power cable.
2. Take ESD precautions by using an anti-static wrist strap.
3. Open the RAM slot clips.
4. Carefully remove the RAM module by holding its edges.
5. Align the notch on the RAM with the slot.
6. Insert the RAM and press it down gently.
7. Make sure both clips lock and the RAM is seated properly.

Storage Installation

1. Shut down the computer and unplug the power cable.
2. Disconnect the SATA data cable and power cable.
3. Remove the storage device from its bay.

4. Place the storage device back into the bay and secure it.
5. Reconnect the SATA data and power cables.
6. Start the computer.
7. Check BIOS/Windows to confirm that the RAM and storage device are detected correctly.

Storage connection type: SATA.

Task 4: Full Desktop PC Assembly



Steps

1. Prepare the workspace and keep all PC components ready.
2. Turn off and unplug the power supply before starting.
3. Take ESD precautions by using an anti-static wrist strap.
4. Install the CPU carefully into the motherboard socket.
5. Install the CPU cooler and connect its fan cable.
6. Install the RAM modules into the correct motherboard slots.

7. Install the storage device (SATA SSD/HDD or NVMe SSD).
8. Place the motherboard inside the PC case and secure it with screws.
9. Install the Power Supply Unit (PSU) in the case.
10. Install the GPU into the appropriate PCIe slot and secure it.
11. Connect the required power, storage, front-panel, and fan cables.
12. Check all components and cable connections before powering on.
13. Connect the monitor, keyboard, and power cable.
14. Power on the PC.
15. Enter BIOS/UEFI and confirm that the CPU, RAM, storage, and GPU are detected.
16. Confirm that the system completes POST successfully and starts normally.

Section C — Mini Capstone Project

C1. First three checks

SCENARIO

ABC Solutions has newly assembled computers that fail to start or show hardware detection errors.

QUESTION

Identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.

ANSWER: -

First I would check the PSU and power connections because incorrect or missing power can cause no lights, no fans, or no display. Second, I would check the RAM because incorrectly seated or unsupported RAM commonly prevents POST. Third I would check the motherboard and CPU installation because socket, BIOS, CPU-power, bent-pin, or cooler problems can stop the system from starting. If the PC powers on but has no picture, I would also check the GPU and monitor connection.

C2. Motherboard–CPU compatibility

SCENARIO

ABC Solutions is diagnosing newly assembled workstations.

QUESTION

Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system you assembled or inspected.

ANSWER:-

The CPU must use the same socket as the motherboard and must be supported by its chipset and BIOS. If the socket differs, the CPU cannot be installed. If the socket matches but the BIOS is too old, the system may power on but fail POST. My inspected system details are: motherboard [insert model], CPU [insert model], socket [insert socket], BIOS [insert version]. I checked the manufacturer support list and recorded whether the CPU was supported or required a BIOS update.

- C3. Verifying RAM and storage

SCENARIO ABC Solutions needs installation verification.

QUESTION

Describe how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.

ANSWER:-

I would check that the RAM notch is aligned, the module is fully inserted, and both retaining clips are locked. I would test one module in the recommended slot if necessary. BIOS should show the expected total RAM. For storage, I would check that an NVMe drive is inserted and secured or that a SATA drive has both data and power cables connected. BIOS should show the drive model and capacity, and the operating system should detect it.

C4. Installation checklist

SCENARIO ABC Solutions wants to prevent similar failures.

QUESTION Produce a hardware-installation checklist of at least three steps.

ANSWER: -

I would check compatibility before assembly; use ESD protection and handle parts by their edges; install CPU, RAM, storage, and cooler correctly; connect the 24-pin and CPU power cables; connect CPU_FAN; seat the GPU and its power cable; inspect for loose screws; test POST and BIOS detection before installing software; and record component models and results.

C5. SSD justification

SCENARIO ABC Solutions is building modern office workstations for the lab.

QUESTION

Provide a technical justification for using SSDs instead of HDDs in modern office workstations, tying it back to the lab use case.

ANSWER:-

SSDs use flash memory and have no moving parts, so they provide faster access, quicker booting, and faster application loading, less noise, lower power use, and better resistance to movement than HDDs. These benefits suit a lab where many users start computers, open office applications, and work with files every day. NVMe SSDs provide the highest performance when supported; SATA SSDs provide strong office performance at a lower cost. HDDs may be retained for large secondary storage or backups.

Section D — AI-Augmented Learning

Step 1: Exact prompt given to AI

Create a simple-English, step-by-step troubleshooting checklist for a junior technician diagnosing a desktop PC that fails to boot immediately after assembly. Put the checks in the correct priority order. Include ESD and electrical safety, PSU and power connections, motherboard and CPU, RAM, GPU and display, storage, BIOS/POST, and the minimum hardware test. Apply the checklist to the ABC Solutions computer-lab scenario, where some newly assembled computers fail to start and others show hardware-detection errors. Explain which step would identify each likely problem. Do not assume exact component models.

AI's original checklist

1. Check that the monitor and keyboard are connected.
2. Turn on the PSU and press the computer's power button.
3. Check that the PSU has enough wattage.
4. Check that the motherboard is compatible with the CPU.
5. Check that the RAM is installed correctly.
6. Check that the storage device is connected.
7. Check the GPU and its power cable.
8. Check BIOS settings and error messages.
9. Replace components that may be faulty.
10. Reinstall the operating system if the PC still does not boot.

Corrected checklist

11. Make the system safe: switch off, unplug, and use ESD precautions.
12. Check wall power, power cable, PSU switch, and surge protector.
13. Check 24-pin motherboard, 4/8-pin CPU, GPU, and storage power cables.
14. Check motherboard/CPU socket compatibility, CPU seating, BIOS support, cooler, CPU_FAN, and bent pins.
15. Reseat RAM and test one supported module in the recommended slot.
16. Reseat GPU, connect its power, and check the correct monitor output.
17. Read motherboard LEDs, beep codes, display messages, and fan behavior.
18. Test minimum hardware: motherboard, CPU, cooler, one RAM module, PSU, and display.

19. Enter BIOS and confirm CPU, RAM, and storage detection; clear CMOS or update BIOS if appropriate.
20. Check SATA/M.2 installation, boot order, and storage health.
21. Test one suspected component at a time with known-good compatible parts.
22. Document the fault, repair, and final POST result.

ABC Solutions application: No power points to steps 2–3; failure to POST points to steps 4–5; missing storage points to steps 9–10; and no display points to step 6.

Step 2: Test & Debug note

The original checklist did not put ESD and electrical safety first, so I added that step before opening the case. I moved internal power checks earlier because the system must receive correct power before other troubleshooting. I added a minimum-hardware test and clearer BIOS/storage checks to isolate one component at a time. I removed operating-system reinstallation from the early process because hardware POST must be confirmed before software troubleshooting.